Exp. No: 7

Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant

Aim:

To determine and plot the capacitance of an isolated sphere and a concentric spherical capacitor for different radii and dielectric constants using MATLAB

Objectives:

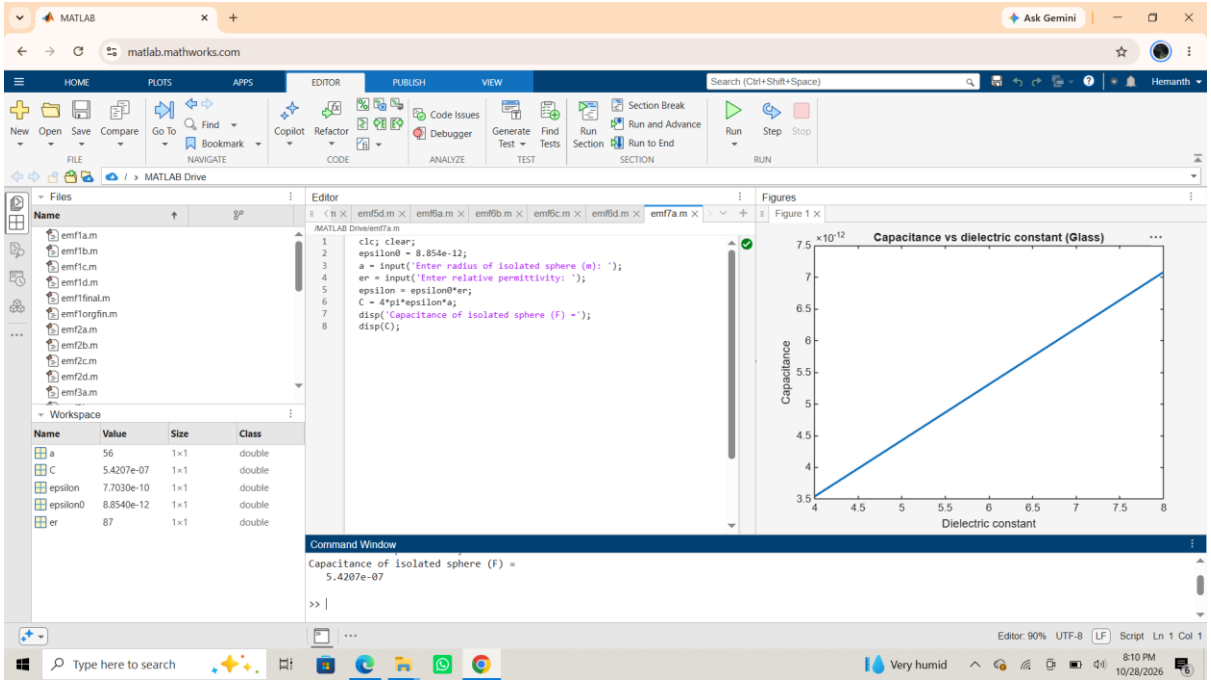
1. Determine the capacitance of an isolated sphere for a specified radius and dielectric constant.
2. Determine the capacitance of a concentric spherical capacitor for specified inner and outer radii and dielectric constant.
3. Investigate the effect of inner radius, outer radius and relative permittivity on capacitance.
4. Compare MATLAB results with the analytical expressions and interpret the observed trends.

Test Case (i) – Capacitance of an Isolated Sphere

MATLAB Program:

```
clc;
clear;
epsilon0 = 8.854e-12;
a = input('Enter radius of isolated sphere (m): ');
er = input('Enter relative permittivity: ');
epsilon = epsilon0*er;
C = 4*pi*epsilon*a;
disp('Capacitance of isolated sphere (F) =');
disp(C);
```

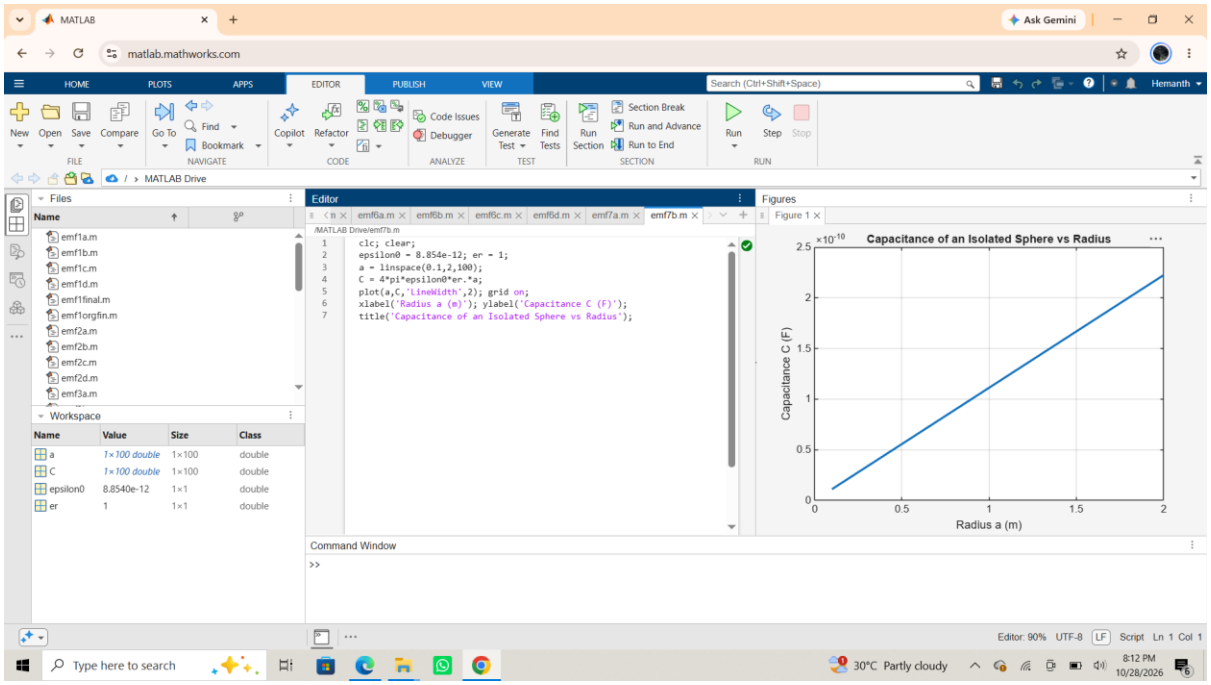
OUTPUT:



Test Case (ii) – Variation of Isolated-Sphere Capacitance with Radius

```
clc;
clear;
epsilon0 = 8.854e-12;
er = 1;
a = linspace(0.1,2,100);
C = 4*pi*epsilon0*er.*a;
plot(a,C,'LineWidth',2); grid on;
xlabel('Radius a (m)'); ylabel('Capacitance C (F)');
title('Capacitance of an Isolated Sphere vs Radius');
```

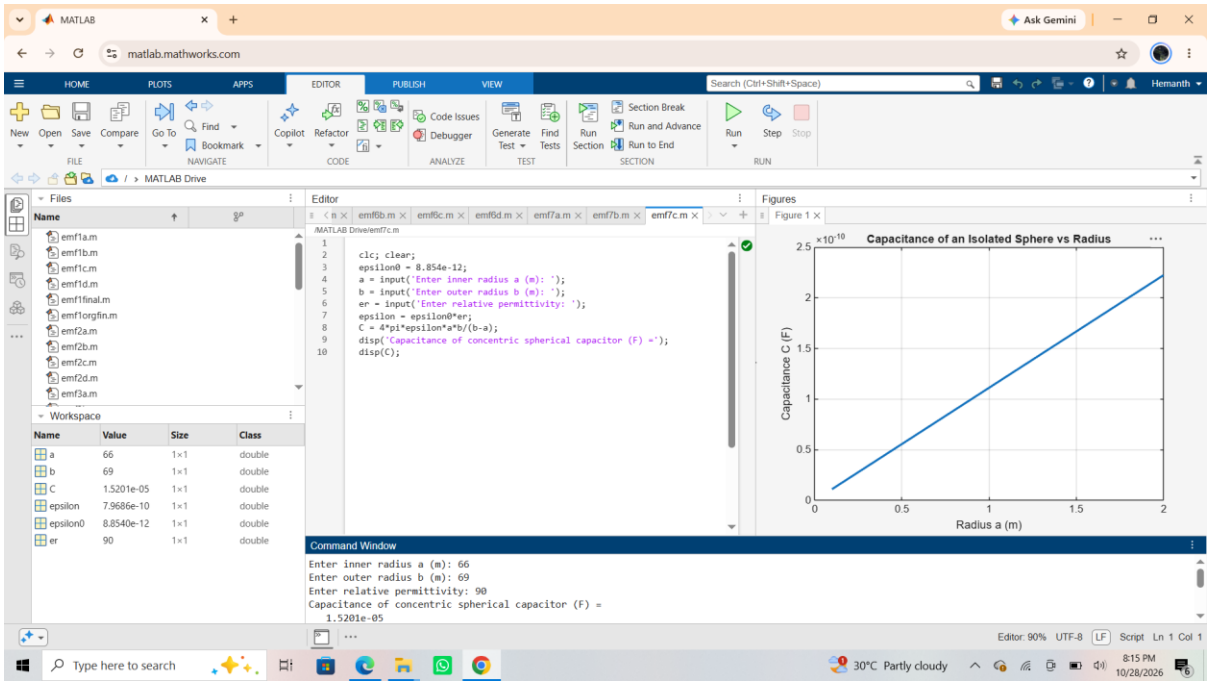
OUTPUT:



Test Case (iii) – Capacitance of a Concentric Spherical Capacitor

```
clc;
 clear;
 epsilon0 = 8.854e-12;
a = input('Enter inner radius a (m): ');
 b = input('Enter outer radius b (m): ');
er = input('Enter relative permittivity: ');
 epsilon = epsilon0*er;
 C = 4*pi*epsilon*a*b/(b-a);
disp('Capacitance of concentric spherical capacitor (F) =');
disp(C);
```

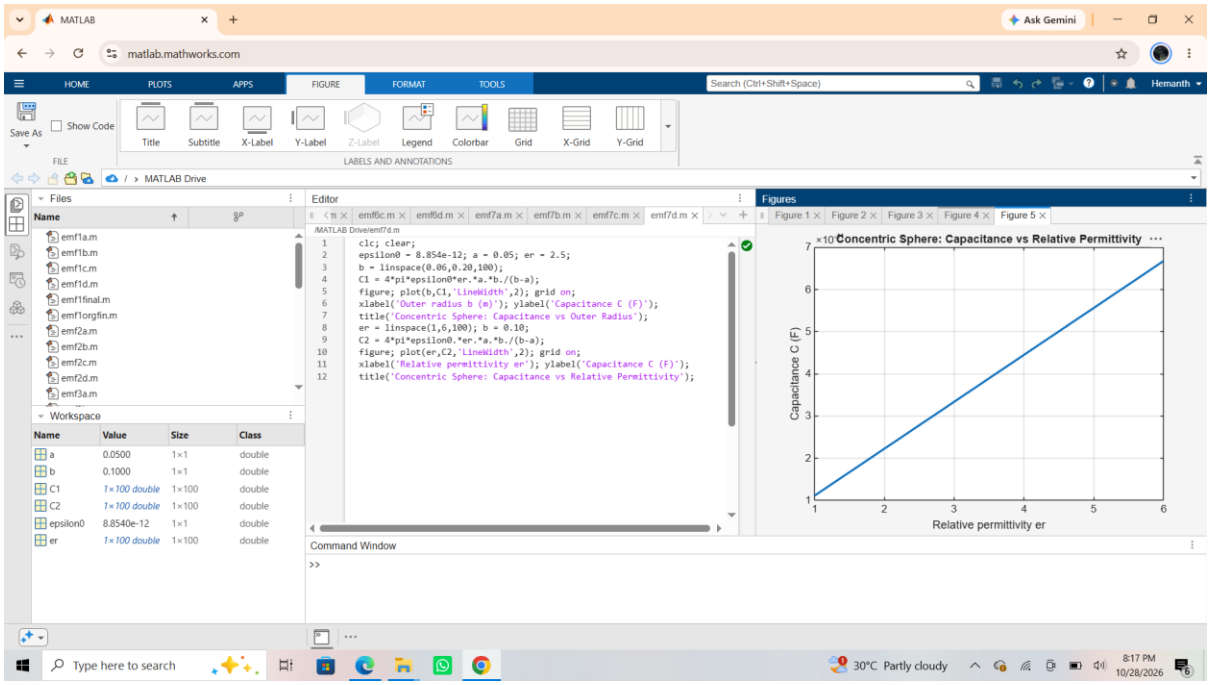
OUTPUT:



Test Case (iv) – Effect of Outer Radius and Relative Permittivity

```
clc;
clear;
epsilon0 = 8.854e-12; a = 0.05; er = 2.5;
b = linspace(0.06,0.20,100);
C1 = 4*pi*epsilon0*er.*a.*b./(b-a);
figure; plot(b,C1,'LineWidth',2); grid on;
xlabel('Outer radius b (m)'); ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Outer Radius');
er = linspace(1,6,100); b = 0.10;
C2 = 4*pi*epsilon0.*er.*a.*b./(b-a);
figure; plot(er,C2,'LineWidth',2); grid on;
xlabel('Relative permittivity er'); ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Relative Permittivity');
```

OUTPUT:



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Define spherical geometry and dielectric	Define spherical geometry and dielectric	. Correctly identifies isolated sphere radius or inner/outer radii and medium permittivity	. Geometry is correct with minor omission.	Some dimensions or material properties are unclear.	. Uses invalid geometry or material input	25	2.5	
2	Implement spherical capacitance equations	Implement spherical capacitance equations	. Applies the appropriate isolated/con centric sphere formulas correctly.	. Formulas are mostly correct with minor errors.	. Partial calculations require guidance.	Uses incorrect equations.	25	2.5	
3	Analyze radius and permittivity effects	Analyze radius and permittivity effects	Compares cases and validates the expected capacitance trends against theory.	Analysis is mostly correct with minor omissions.	Comparisons or validation are incomplete.	No valid trend analysis is shown.	30	3	

4	Present results and conclusions	Present results and conclusions	Reports geometry, units, calculations	Report is clear with minor gaps.	Report lacks some key information	Results are unclear or missing.	20	2	